Министерство науки и высшего образования Российской Федерации
ФЕДЕРАЛЬНОЕ ГОСУДАРСТВЕННОЕ АВТОНОМНОЕ ОБРАЗОВАТЕЛЬНОЕ УЧРЕЖДЕНИЕ ВЫСШЕГО ОБРАЗОВАНИЯ
«Национальный исследовательский университет ИТМО»
(Университет ИТМО)

Факультет прикладной информатики
Направление подготовки 09.03.03 Прикладная информатика
Образовательная программа Мобильные и сетевые технологии

О Т Ч Е Т

по лабораторной работе по предмету Проектирование и реализация баз данных №3.2
«Создание таблиц базы данных PostgreSQL. Заполнение таблиц рабочими данными»

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Цель работы: овладеть практическими навыками создания таблиц базы данных PostgreSQL 1X, заполнения их рабочими данными, резервного копирования и восстановления БД.

Практическое задание:

1. Создать базу данных с использованием pgAdmin 4 (согласно индивидуальному заданию).
2. Создать схему в составе базы данных.
3. Создать таблицы базы данных.
4. Установить ограничения на данные: Primary Key, Unique, Check, Foreign Key.
5. Заполнить таблицы БД рабочими данными.
6. Создать резервную копию БД.

Указание:

Создать две резервные копии:

- с расширением CUSTOM для восстановления БД;
- с расширением для листинга (в отчете);
- при создании резервных копий БД настроить параметры Dump options для Type of objects и Queries .

7. Восстановить БД.

Вариант 4: БД “Учет выполнения заданий”

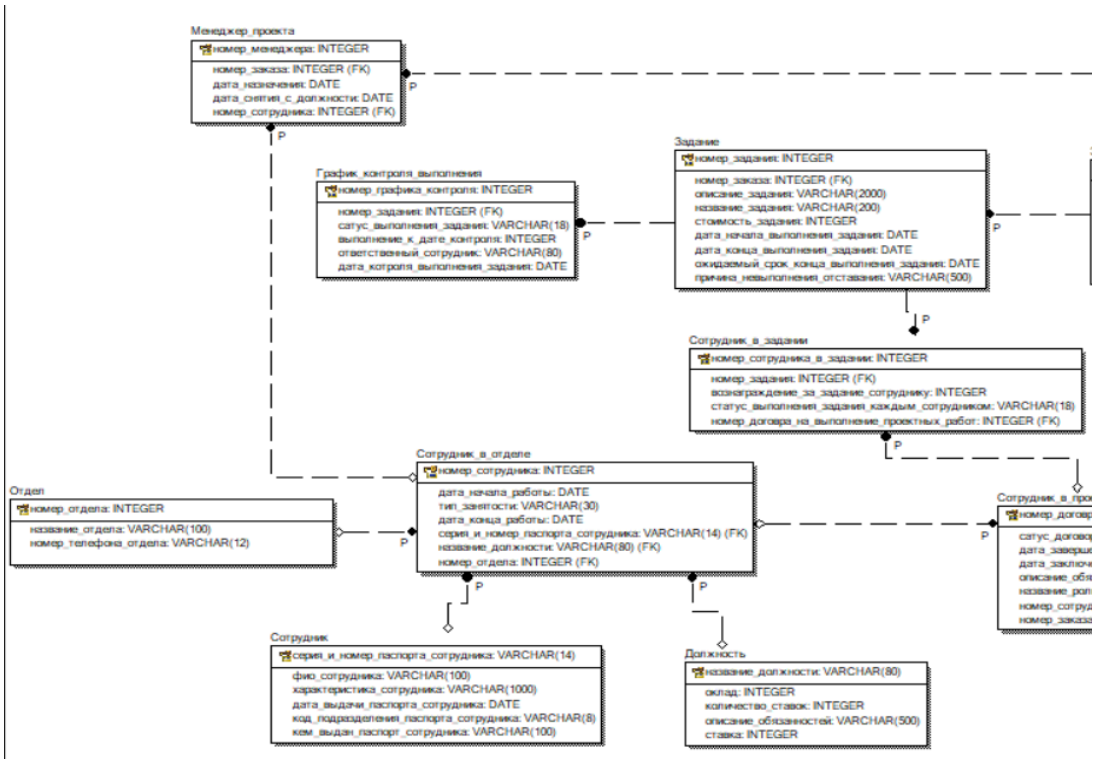

Рисунок 1.1 - Схема инфологической модели БД ЛР 2 (IDEF1X) 1 половина

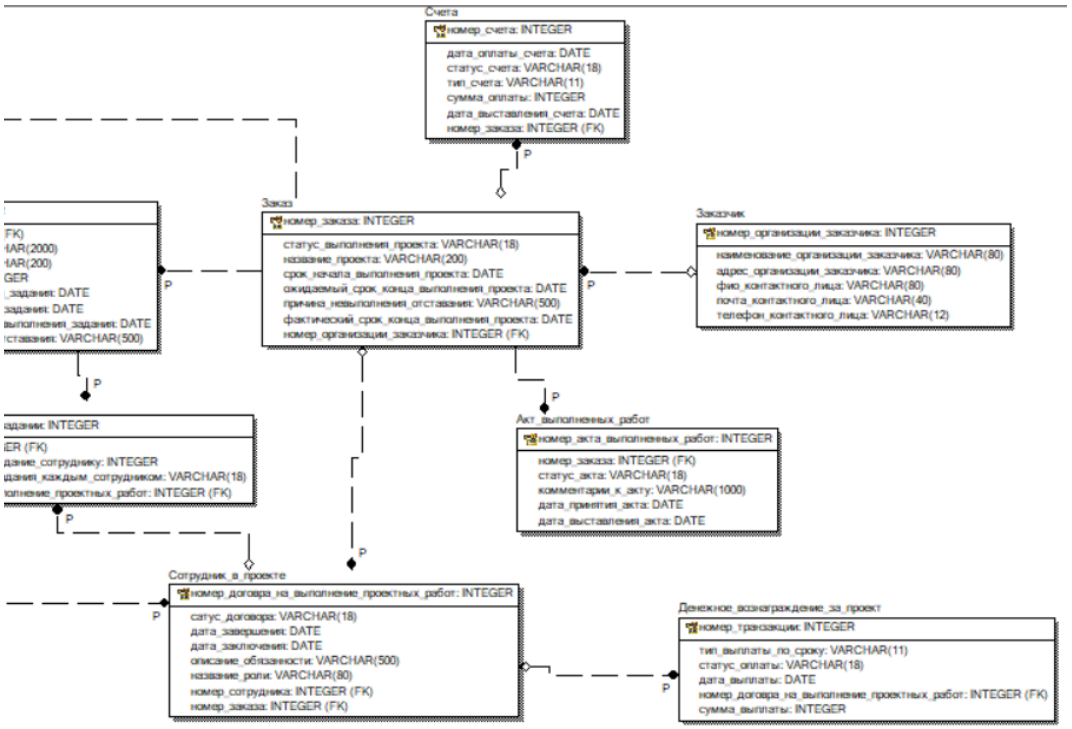

Рисунок 1.2 - Схема инфологической модели БД ЛР 2 (IDEF1X) 2 половина

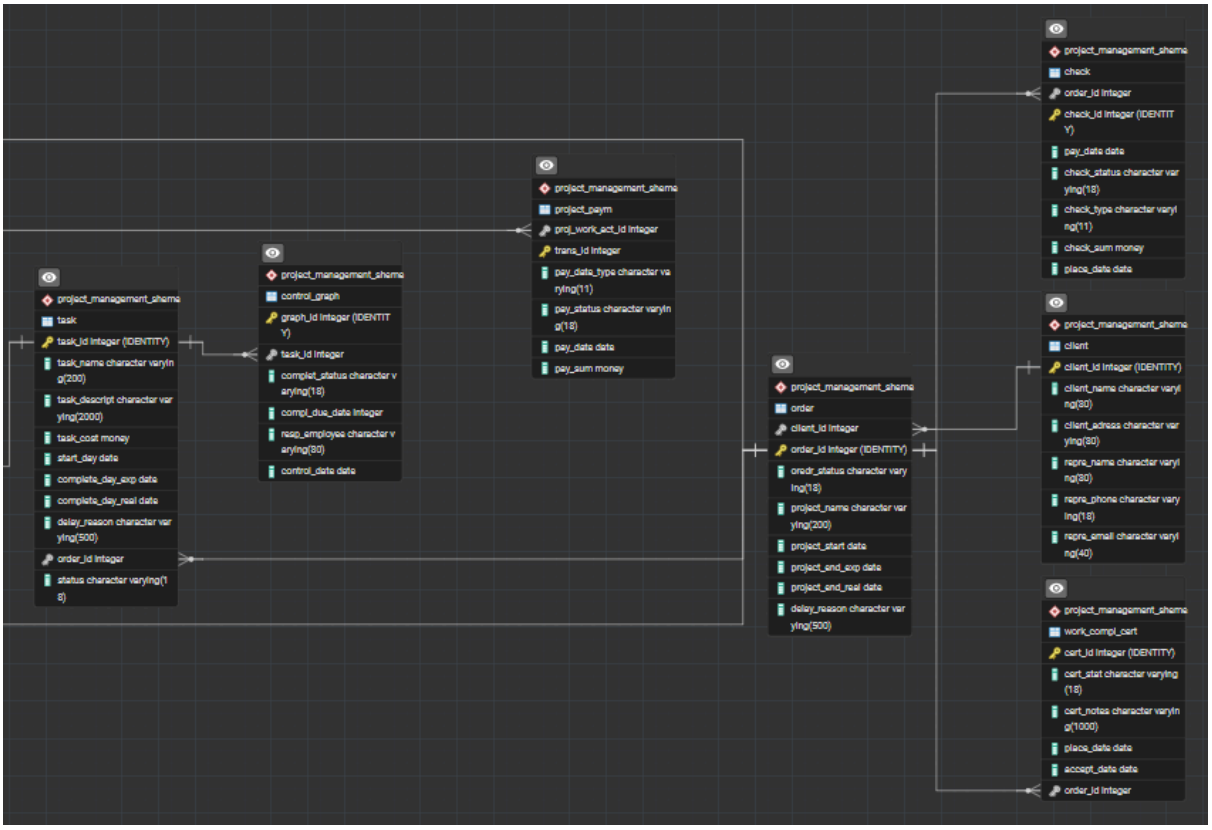

Рисунок 2.1 - Схема логической модели базы данных Generate ERD 1 половина

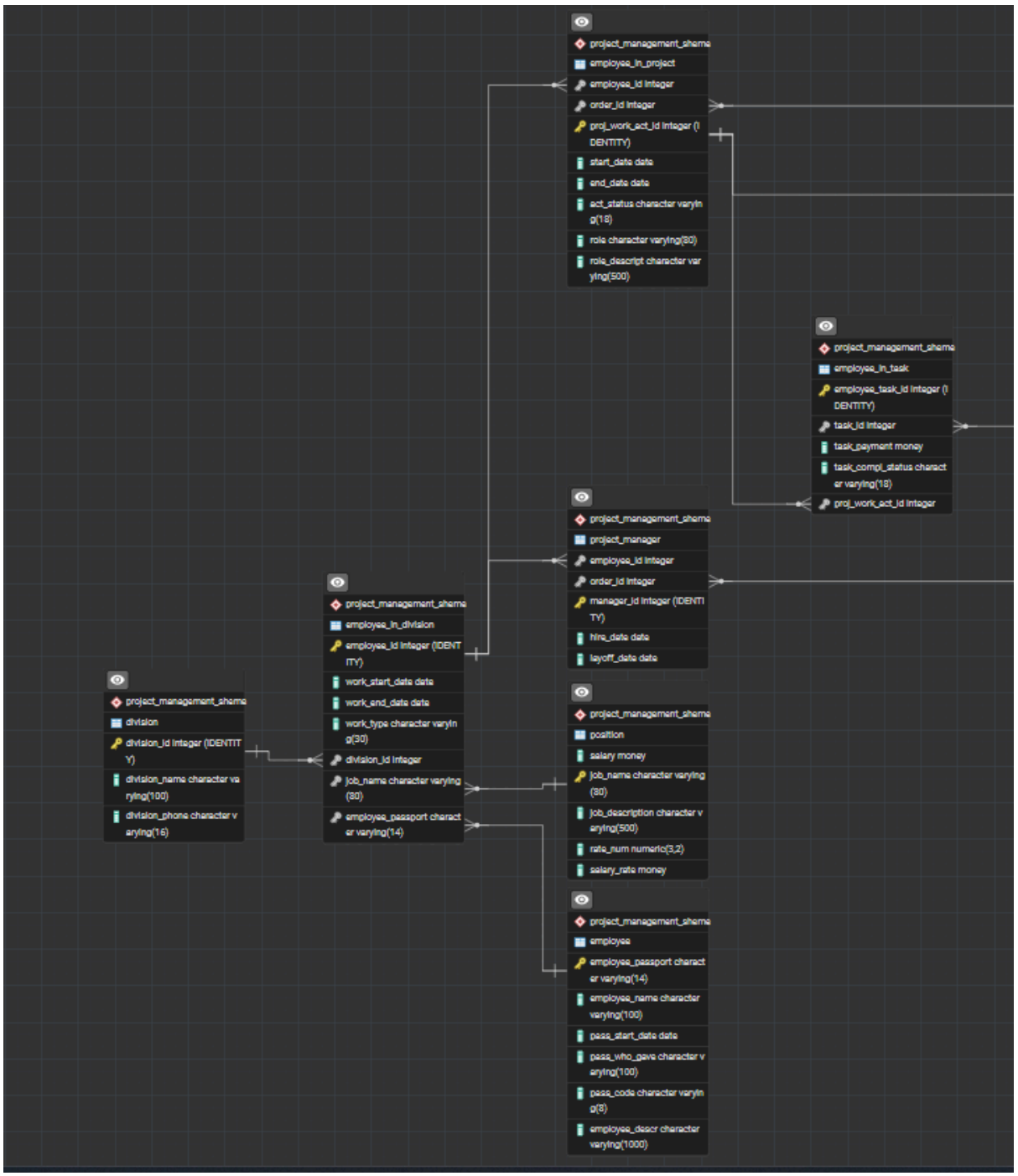

Рисунок 2.2 - Схема логической модели базы данных Generate ERD 2 половина

Дамп БД (частично - полноценная версия отдельным файлом):

--

-- PostgreSQL database dump

--

\restrict
t2thrlKuskox5mcjl3xqGrRMBVjS5Uyzf0r4htyvWs1b1tbhPSQaq9FEidIHjaO

-- Dumped from database version 13.23

-- Dumped by pg_dump version 13.23

-- Started on 2026-04-10 15:07:10 // дата дампа

SET statement_timeout = 0;

SET lock_timeout = 0;

SET idle_in_transaction_session_timeout = 0;

SET client_encoding = 'UTF8';

SET standard_conforming_strings = on;

SELECT pg_catalog.set_config('search_path', '', false);

SET check_function_bodies = false;

SET xmloption = content;

SET client_min_messages = warning;

SET row_security = off;

--

-- TOC entry 6 (class 2615 OID 16395)

-- Name: project_management_sheme; Type: SCHEMA; Schema: -; Owner: postgres

--

```
CREATE SCHEMA project_management_sheme;



ALTER SCHEMA project_management_sheme OWNER TO postgres;


SET default_tablespace = '';


SET default_table_access_method = heap;


--

-- TOC entry 204 (class 1259 OID 16446)

-- Name: check; Type: TABLE; Schema: project_management_sheme;
Owner: postgres                    // Имя БД

--

// Создание таблиц

CREATE TABLE project_management_sheme."check" (

    order_id integer NOT NULL,

    check_id integer NOT NULL,

    pay_date date,

    check_status character varying(18) NOT NULL,

    check_type character varying(11) NOT NULL,

    place_date date NOT NULL,

    check_sum numeric(15,2) NOT NULL

);
```

```
ALTER TABLE project_management_sheme."check" OWNER TO postgres;


--

-- TOC entry 222 (class 1259 OID 16663)

-- Name: check_check_id_seq; Type: SEQUENCE; Schema:
project_management_sheme; Owner: postgres

--


ALTER TABLE project_management_sheme."check" ALTER COLUMN
check_id ADD GENERATED ALWAYS AS IDENTITY (

    SEQUENCE NAME project_management_sheme.check_check_id_seq

    START WITH 1

    INCREMENT BY 1

    NO MINVALUE

    NO MAXVALUE

    CACHE 1

);




--

-- TOC entry 202 (class 1259 OID 16433)

-- Name: client; Type: TABLE; Schema: project_management_sheme;
Owner: postgres

--
```

```
CREATE TABLE project_management_sheme.client (
    client_id integer NOT NULL,
    client_name character varying(80) NOT NULL,
    client_adress character varying(80) NOT NULL,
    repre_name character varying(80) NOT NULL,
    repre_phone character varying(18) NOT NULL,
    repre_email character varying(40)
);


ALTER TABLE project_management_sheme.client OWNER TO postgres;


--
-- TOC entry 216 (class 1259 OID 16634)
-- Name: client_client_id_seq; Type: SEQUENCE; Schema: project_management_sheme; Owner: postgres
--


ALTER TABLE project_management_sheme.client ALTER COLUMN client_id ADD GENERATED ALWAYS AS IDENTITY (
    SEQUENCE NAME project_management_sheme.client_client_id_seq
    START WITH 1
    INCREMENT BY 1
    NO MINVALUE
    NO MAXVALUE
```

```
    CACHE 10
);



--
-- TOC entry 213 (class 1259 OID 16600)
-- Name: control_graph; Type: TABLE; Schema:
project_management_sheme; Owner: postgres
--

CREATE TABLE project_management_sheme.control_graph (
    graph_id integer NOT NULL,
    task_id integer NOT NULL,
    complet_status character varying(18) NOT NULL,
    compl_due_date integer,
    resp_employee character varying(80) NOT NULL,
    control_date date NOT NULL
);



ALTER TABLE project_management_sheme.control_graph OWNER TO
postgres;


--
-- TOC entry 223 (class 1259 OID 16670)
```

```
-- Name: control_graph_graph_id_seq; Type: SEQUENCE; Schema: project_management_sheme; Owner: postgres

--


ALTER TABLE project_management_sheme.control_graph ALTER COLUMN graph_id ADD GENERATED ALWAYS AS IDENTITY (

    SEQUENCE NAME project_management_sheme.control_graph_graph_id_seq

    START WITH 1

    INCREMENT BY 1

    NO MINVALUE

    NO MAXVALUE

    CACHE 1

);




--

-- TOC entry 206 (class 1259 OID 16466)

-- Name: division; Type: TABLE; Schema: project_management_sheme; Owner: postgres

--


CREATE TABLE project_management_sheme.division (

    division_id integer NOT NULL,

    division_name character varying(100) NOT NULL,

    division_phone character varying(18) NOT NULL
```

```
);



ALTER TABLE project_management_sheme.division OWNER TO postgres;



--

-- TOC entry 220 (class 1259 OID 16656)

-- Name: division_division_id_seq; Type: SEQUENCE; Schema:
project_management_sheme; Owner: postgres

--


ALTER TABLE project_management_sheme.division ALTER COLUMN
division_id ADD GENERATED ALWAYS AS IDENTITY (

    SEQUENCE NAME
project_management_sheme.division_division_id_seq

    START WITH 1

    INCREMENT BY 1

    NO MINVALUE

    NO MAXVALUE

    CACHE 1

);



--

-- TOC entry 207 (class 1259 OID 16471)
```

```
-- Name: employee; Type: TABLE; Schema: project_management_sheme;
Owner: postgres

--


CREATE TABLE project_management_sheme.employee (

    employee_passport character varying(14) NOT NULL,

    employee_name character varying(100) NOT NULL,

    pass_start_date date NOT NULL,

    pass_who_gave character varying(100) NOT NULL,

    pass_code character varying(8) NOT NULL,

    employee_descr character varying(1000)

);



ALTER TABLE project_management_sheme.employee OWNER TO
postgres;


--

-- TOC entry 208 (class 1259 OID 16479)

-- Name: employee_in_division; Type: TABLE; Schema:
project_management_sheme; Owner: postgres

--


CREATE TABLE project_management_sheme.employee_in_division (

    employee_id integer NOT NULL,

    work_start_date date NOT NULL,
```

```
    work_end_date date,
    work_type character varying(30) NOT NULL,
    division_id integer NOT NULL,
    job_name character varying(80) NOT NULL,
    employee_passport character varying(14) NOT NULL
);



ALTER TABLE project_management_sheme.employee_in_division
OWNER TO postgres;



--
-- TOC entry 218 (class 1259 OID 16642)
-- Name: employee_in_division_employee_id_seq; Type: SEQUENCE;
Schema: project_management_sheme; Owner: postgres
--


ALTER TABLE project_management_sheme.employee_in_division ALTER
COLUMN employee_id ADD GENERATED ALWAYS AS IDENTITY (
    SEQUENCE NAME
project_management_sheme.employee_in_division_employee_id_seq
    START WITH 1
    INCREMENT BY 1
    NO MINVALUE
    NO MAXVALUE
    CACHE 1
```

```
);



--
-- TOC entry 210 (class 1259 OID 16505)
-- Name: employee_in_project; Type: TABLE; Schema:
project_management_sheme; Owner: postgres
--

CREATE TABLE project_management_sheme.employee_in_project (
    employee_id integer NOT NULL,
    order_id integer NOT NULL,
    proj_work_act_id integer NOT NULL,
    start_date date,
    end_date date,
    act_status character varying(18) NOT NULL,
    role character varying(80) NOT NULL,
    role_descript character varying(500) NOT NULL
);



ALTER TABLE project_management_sheme.employee_in_project OWNER
TO postgres;


--
-- TOC entry 219 (class 1259 OID 16652)
```

-- Name: employee_in_project_proj_work_act_id_seq; Type: SEQUENCE; Schema: project_management_sheme; Owner: postgres

--

ALTER TABLE project_management_sheme.employee_in_project ALTER COLUMN proj_work_act_id ADD GENERATED ALWAYS AS IDENTITY (

    SEQUENCE NAME project_management_sheme.employee_in_project_proj_work_act_id_seq

    START WITH 1

    INCREMENT BY 1

    NO MINVALUE

    NO MAXVALUE

    CACHE 1

);

--

-- TOC entry 214 (class 1259 OID 16611)

-- Name: employee_in_task; Type: TABLE; Schema: project_management_sheme; Owner: postgres

--

CREATE TABLE project_management_sheme.employee_in_task (

    employee_task_id integer NOT NULL,

    task_id integer,

    task_payment money,

```
    task_compl_status character varying(18) NOT NULL,

    proj_work_act_id integer

);



ALTER TABLE project_management_sheme.employee_in_task OWNER TO
postgres;



--

-- TOC entry 225 (class 1259 OID 16678)

-- Name: employee_in_task_employee_task_id_seq; Type: SEQUENCE;
Schema: project_management_sheme; Owner: postgres

--


ALTER TABLE project_management_sheme.employee_in_task ALTER
COLUMN employee_task_id ADD GENERATED ALWAYS AS IDENTITY
(

    SEQUENCE NAME
project_management_sheme.employee_in_task_employee_task_id_seq

    START WITH 1

    INCREMENT BY 1

    NO MINVALUE

    NO MAXVALUE

    CACHE 1

);
```

```
--

-- TOC entry 203 (class 1259 OID 16438)

-- Name: order; Type: TABLE; Schema: project_management_sheme;
Owner: postgres

--


CREATE TABLE project_management_sheme."order" (

    client_id integer NOT NULL,

    order_id integer NOT NULL,

    oredr_status character varying(18) NOT NULL,

    project_name character varying(200) NOT NULL,

    project_start date NOT NULL,

    project_end_exp date NOT NULL,

    project_end_real date,

    delay_reason character varying(500)

);



ALTER TABLE project_management_sheme."order" OWNER TO postgres;


--

-- TOC entry 215 (class 1259 OID 16626)

-- Name: order_order_id_seq; Type: SEQUENCE; Schema:
project_management_sheme; Owner: postgres

--
```

```
ALTER TABLE project_management_sheme."order" ALTER COLUMN
order_id ADD GENERATED ALWAYS AS IDENTITY (

    SEQUENCE NAME project_management_sheme.order_order_id_seq

    START WITH 1

    INCREMENT BY 1

    NO MINVALUE

    NO MAXVALUE

    CACHE 1

);




--

-- TOC entry 201 (class 1259 OID 16413)

-- Name: position; Type: TABLE; Schema: project_management_sheme;
Owner: postgres

--


CREATE TABLE project_management_sheme."position" (

    salary numeric(10,2) DEFAULT 0 NOT NULL,

    job_name character varying(80) DEFAULT 'Undefined'::character varying
NOT NULL,

    job_description character varying(500) DEFAULT 'none'::character
varying NOT NULL,

    rate_num numeric(3,2) DEFAULT 1 NOT NULL,

    salary_rate numeric GENERATED ALWAYS AS
((((salary)::numeric)::double precision / (NULLIF((rate_num *
(40)::numeric), (0)::numeric))::double precision)) STORED
```

```
);



ALTER TABLE project_management_sheme."position" OWNER TO
postgres;



--

-- TOC entry 209 (class 1259 OID 16492)

-- Name: project_manager; Type: TABLE; Schema:
project_management_sheme; Owner: postgres

--


CREATE TABLE project_management_sheme.project_manager (

    employee_id integer NOT NULL,

    order_id integer NOT NULL,

    manager_id integer NOT NULL,

    hire_date date NOT NULL,

    layoff_date date

);



ALTER TABLE project_management_sheme.project_manager OWNER TO
postgres;



--

-- TOC entry 224 (class 1259 OID 16674)
```

-- Name: project_manager_manager_id_seq; Type: SEQUENCE; Schema: project_management_sheme; Owner: postgres

--

ALTER TABLE project_management_sheme.project_manager ALTER COLUMN manager_id ADD GENERATED ALWAYS AS IDENTITY (

    SEQUENCE NAME project_management_sheme.project_manager_manager_id_seq

    START WITH 1

    INCREMENT BY 1

    NO MINVALUE

    NO MAXVALUE

    CACHE 1

);

--

-- TOC entry 211 (class 1259 OID 16519)

-- Name: project_paym; Type: TABLE; Schema: project_management_sheme; Owner: postgres

--

CREATE TABLE project_management_sheme.project_paym (

    proj_work_act_id integer NOT NULL,

    trans_id integer NOT NULL,

    pay_date_type character varying(11) NOT NULL,

```
    pay_status character varying(18) NOT NULL,
    pay_date date,
    pay_sum money NOT NULL
);



ALTER TABLE project_management_sheme.project_paym OWNER TO
postgres;


--
-- TOC entry 212 (class 1259 OID 16532)
-- Name: task; Type: TABLE; Schema: project_management_sheme; Owner:
postgres
--


CREATE TABLE project_management_sheme.task (
    task_id integer NOT NULL,
    task_name character varying(200) NOT NULL,
    task_descript character varying(2000),
    task_cost numeric(15,2) NOT NULL,
    start_day date NOT NULL,
    complete_day_exp date NOT NULL,
    complete_day_real date,
    delay_reason character varying(500),
    order_id integer NOT NULL,
    status character varying(18) NOT NULL
```

```
);



ALTER TABLE project_management_sheme.task OWNER TO postgres;



--

-- TOC entry 217 (class 1259 OID 16636)

-- Name: task_task_id_seq; Type: SEQUENCE; Schema:
project_management_sheme; Owner: postgres

--


ALTER TABLE project_management_sheme.task ALTER COLUMN task_id
ADD GENERATED ALWAYS AS IDENTITY (

    SEQUENCE NAME project_management_sheme.task_task_id_seq

    START WITH 1

    INCREMENT BY 1

    NO MINVALUE

    NO MAXVALUE

    CACHE 1

);



--

-- TOC entry 205 (class 1259 OID 16458)

-- Name: work_compl_cert; Type: TABLE; Schema:
project_management_sheme; Owner: postgres
```

```
--

CREATE TABLE project_management_sheme.work_compl_cert (
    cert_id integer NOT NULL,
    cert_stat character varying(18) NOT NULL,
    cert_notes character varying(1000),
    place_date date NOT NULL,
    accept_date date,
    order_id integer NOT NULL
);


ALTER TABLE project_management_sheme.work_compl_cert OWNER TO postgres;


--
-- TOC entry 221 (class 1259 OID 16658)
-- Name: work_compl_cert_cert_id_seq; Type: SEQUENCE; Schema: project_management_sheme; Owner: postgres
--

ALTER TABLE project_management_sheme.work_compl_cert ALTER COLUMN cert_id ADD GENERATED ALWAYS AS IDENTITY (
    SEQUENCE NAME project_management_sheme.work_compl_cert_cert_id_seq
    START WITH 1
```

```
    INCREMENT BY 1
    NO MINVALUE
    NO MAXVALUE
    CACHE 1
);



--
-- TOC entry 3154 (class 0 OID 16446)
-- Dependencies: 204
-- Data for Name: check; Type: TABLE DATA; Schema:
project_management_sheme; Owner: postgres
--
// Заполнение данными - пропущено, так как много строк
COPY project_management_sheme."check" (order_id, check_id, pay_date,
check_status, check_type, place_date, check_sum) FROM stdin;
1	1	\N	отклонен	остаток	2025-08-18	74422629054.00
2	2	\N	выставлен	остаток	2025-09-01	98621687833.00
3	3	\N	выставлен	предоплата	2026-03-26	1117892422.00
4	4	\N	отклонен	остаток	2025-05-01	27933066568.00
5	5	\N	выставлен	остаток	2026-04-30	25107862667.00
…
97	выставлен	ed232cea9aff4335b607c2537371df2b	2025-05-23	\N	97
98	выставлен	e5f24cda09857a14d643b8585fcd8128	2025-12-01	\N	98
99	выставлен	37171f6ffb51406b235e31c853be119e	2026-01-19	\N	99
```

100выставлен 78622665875d4eb9d8dda9ab9ec64f5a 2025-10-22 \N 100

\.

-- Completed on 2026-04-10 15:07:11

--

-- PostgreSQL database dump complete

--

\unrestrict t2thrlKuskox5mcjl3xqGrRMBVjS5Uyzf0r4htyvWs1b1tbhPSQaq9FEidIHjaO

Выводы: Мы успешно создали базу данных и наполнили ее тестовыми данными для данной предметной области